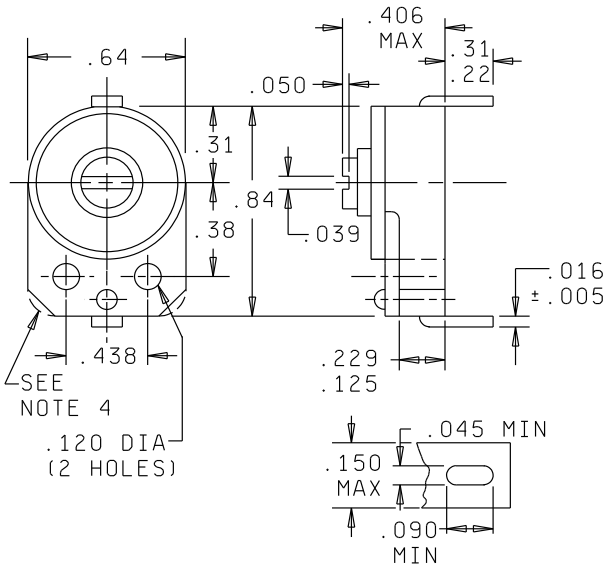
INCH-POUND
MIL-PRF-81/1F
24 April 2017
SUPERSEDING
MIL-PRF-81/1E
19 April 2011

PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, VARIABLE, CERAMIC DIELECTRIC, STYLE CV11

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81.




| Inches | mm |
|---|---|
| .005 | 0.13 |
| .016 | 0.41 |
| .039 | 0.99 |
| .045 | 1.14 |
| .050 | 1.27 |
| .090 | 2.29 |
| .120 | 3.05 |
| .125 | 3.18 |
| .150 | 3.81 |
| .22 | 5.59 |
| .229 | 5.82 |
| .31 | 7.87 |
| .38 | 9.65 |
| .406 | 10.31 |
| .438 | 11.13 |
| .64 | 16.26 |
| .84 | 21.34 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 inch (0.76 mm) for two decimal places and ±.010 inch (0.25 mm) for three decimal places.
4. Limiting dimensions as shown.

FIGURE 1. Dimensions and configurations.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

DC rated voltage: See table I.

Operating temperature range: -55°C to +85°C.

Characteristics: See table I.

Dielectric withstanding voltage (DWV): In accordance with MIL-PRF-81, terminal to terminal.

TABLE I. Style CV11.

| Part or Identifying Number (PIN) | Capacitance (pF) | | DC rated voltage (volts) | Characteristics | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Symbol | Capacitance change from value at +25°C | | | |
| | | | | | At -55°C | | At +85°C | |
| | Minimum | Maximum | | | Minimum (percent) | Maximum (percent) | Minimum (percent) | Maximum (percent) |
| CV11A070 | 1.5 | 7.0 | 500 | A | -4.5 | +2.0 | -2.5 | +2.0 |
| CV11A120 | 3.0 | 12.0 | 500 | A | -4.5 | +2.0 | -2.5 | +2.0 |
| CV11A250 | 4.5 | 25.0 | 500 | A | -4.5 | +2.0 | -2.5 | +2.0 |
| CV11B130 | 3.0 | 13.0 | 500 | B | -1.0 | +3.5 | -2.5 | -0.5 |
| CV11B200 | 5.0 | 20.0 | 500 | B | -1.0 | +3.5 | -2.5 | -0.5 |
| CV11C300 | 4.0 | 30.0 | 500 | C | -1.0 | +6.5 | -4.0 | -1.0 |
| CV11C450 | 7.0 | 45.0 | 500 | C | -1.0 | +6.5 | -4.0 | -1.0 |
| CV11D060 | 2.0 | 6.0 | 500 | D | +1.5 | +8.2 | -5.0 | -1.5 |
| CV11D300 | 4.0 | 30.0 | 500 | D | +1.5 | +8.2 | -5.0 | -1.5 |
| CV11D450 | 7.0 | 45.0 | 500 | D | +1.5 | +8.2 | -5.0 | -1.5 |

Barometric pressure (reduced): In accordance with MIL-PRF-81. The following details shall apply:

Test conditions: D (100,000 feet) and B (50,000 feet) of MIL-STD-202-105.

Test potential: 80 percent of dc rated voltage for condition D and 125 percent dc rated voltage for condition B.

Torque: Not less than 4 ounce-inches nor more than 24 ounce-inches.

Fatigue:

Torque: Not less than 4 ounce-inches nor more than 36 ounce-inches.

Moisture resistance:

Capacitance change: Shall not change more than ±10 percent of initial value or 0.5 pF, whichever is greater.

Qualification inspection: Qualification inspection is not required. A manufacturer that is currently listed on the QPL for at least one other MIL-PRF-81 style may supply CV11 parts by performing the group A and group B testing. A manufacturer that is not qualified to any MIL-PRF-81 styles may produce these parts by performing the group A, group B, and group C testing on the product before delivery.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.

Reference documents: In addition to MIL-PRF-81, this document references the following:

MIL-STD-202-105

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI
Air Force – 19, 99

Preparing activity:
DLA - CC

(Project 5910-2017-016)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/ .



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:40Z
Check the source to verify that this is the current version before use.